



Understanding Data Roles

Data Analyst vs Data Scientist vs AI Engineer

Week 1



Data Analyst **vs** Data Scientist

Both data analysts and data scientists work with **historical data** — but their purpose is different. The data analyst looks at the past to understand what happened and why, while the data scientist looks at the same data to predict what might happen next and recommend what to do about it.

Data Analyst

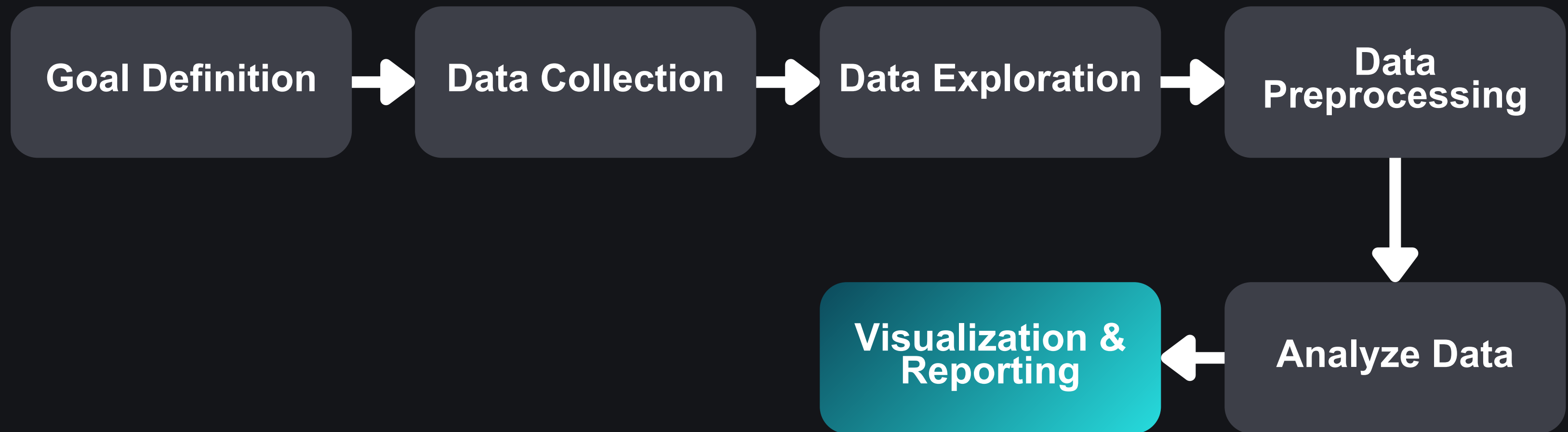
**Describes
what
happened**

Data Scientist

**Predicts
what might
happen**

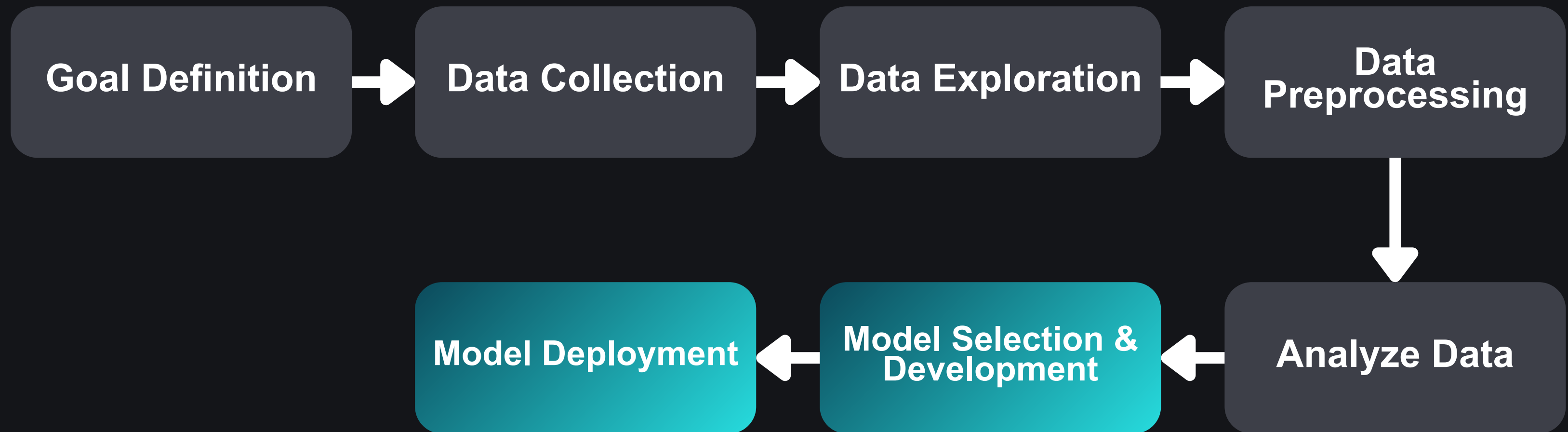


Data Analysis Process





Data Science Process





Comparison

Aspect	Data Analyst	Data Scientist
Data Type	Both work on structured and historical data	
Tools	Both have common tools such as Python, Pandas, R, SQL, etc.	
Goal	Understand what happened and why	Predict what will happen
Usage of Machine Learning	Does not build or train ML models.	Frequently builds and tunes machine learning or deep learning models.
Output	Reports, visualizations	Models, recommendations



Use Case: Ice Cream Dataset

→ Here we have a dummy historical data of an ice cream store:

Flavor	Units Sold	Temperature	Day of Week	Branch
Vanilla	120	33°C	Monday	Cairo
Chocolate	90	33°C	Monday	Giza
Mango	110	35°C	Tuesday	Cairo
...	...	...	...	...

© Use Case: Ice Cream Dataset

Data Analyst's Role:

Goal

Understand past performance and explain trends.

Task

- Analyze which flavors sell the most in each branch
- Discover if they sell more on weekends than on weekdays
- Find the impact of Temperature on sales

Example of the output

→ A report that contains insights such as:

- 1- "Sales peak on Fridays and Saturdays."
- 2- "Cairo branch sells more fruit-based flavors midweek."



Use Case: Ice Cream Dataset

Data Scientist's Role:

Goal

Predict future sales and recommend actions based on patterns

Task

Build a regression model to predict how many units will be sold based on:

- Temperature
- Day of the week
- Flavor
- Branch

Example of the output

→ A machine learning model that can predict the sales of Vanilla Ice cream on Friday in Cairo branch when the temperature is 35°C



What about AI Engineer?

Tasks

To develop AI applications by building and deploying machine learning, computer vision, and natural language processing (NLP) models. (they work with both structured (tables, numbers) and unstructured data (images, text, audio))



Comparison

Role	Focus	Example Output
Data Analyst	Understand past trends	Reports, dashboards
Data Scientist	Predict future outcomes	Trained ML models
AI Engineer	Build & deploy AI apps	AI systems (such as recommendation engine, chatbot)

Thank you

Credits: Ahd Ehab